

Module 4

Circuit Analysis in the S Domain

Dr. Elizabeth Rufus

Professor, School of Electronics Engineering

Laplace Transform pairs

$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s + a}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
te^{-at}	$\frac{1}{(s + a)^2}$
$t^n e^{-at}$	$\frac{n!}{(s + a)^{n+1}}$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$

$f(t)$	$F(s)$
$\sin(\omega t + \theta)$	$\frac{s \sin \theta + \omega \cos \theta}{s^2 + \omega^2}$
$\cos(\omega t + \theta)$	$\frac{s \cos \theta - \omega \sin \theta}{s^2 + \omega^2}$
$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$
$e^{-at} \cos \omega t$	$\frac{s + a}{(s + a)^2 + \omega^2}$

Laplace Transform properties

Linearity: If $F_1(s)$ and $F_2(s)$ are, respectively, the Laplace transforms of $f_1(t)$ and $f_2(t)$, then,

$$\mathcal{L}[a_1 f_1(t) + a_2 f_2(t)] = a_1 F_1(s) + a_2 F_2(s)$$

Where a_1 and a_2 are constants

$$\mathcal{L}[\cos \omega t u(t)] = \mathcal{L}\left[\frac{1}{2}(e^{j\omega t} + e^{-j\omega t})\right] = \frac{1}{2}\mathcal{L}[e^{j\omega t}] + \frac{1}{2}\mathcal{L}[e^{-j\omega t}]$$

$$\mathcal{L}[e^{-at}] = 1/(s + a), \quad \mathcal{L}[\cos \omega t u(t)] = \frac{1}{2}\left(\frac{1}{s - j\omega} + \frac{1}{s + j\omega}\right) = \frac{s}{s^2 + \omega^2}$$

Scaling: If $F(s)$ is the Laplace transform of $f(t)$, then

$$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$$

where a is a constant and $a > 0$

If we let $x = at$, $dx = a dt$, then

$$\mathcal{L}[f(at)] = \int_{0^-}^{\infty} f(x) e^{-x(s/a)} \frac{dx}{a} = \frac{1}{a} \int_{0^-}^{\infty} f(x) e^{-x(s/a)} dx$$

$$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$$

Laplace Transform properties

Time shift: If $F(s)$ is the Laplace transform of $f(t)$, then

$$\mathcal{L}[f(t-a)u(t-a)] = \int_{0^-}^{\infty} f(t-a)u(t-a)e^{-st} dt$$
$$a \geq 0$$

But $u(t-a) = 0$ for $t < a$ and $u(t-a) = 1$ for $t > a$. Hence,

$$\mathcal{L}[f(t-a)u(t-a)] = \int_a^{\infty} f(t-a)e^{-st} dt$$

If we let $x = t - a$, then $dx = dt$ and $t = x + a$. As $t \rightarrow a$, $x \rightarrow 0$ and as $t \rightarrow \infty$, $x \rightarrow \infty$. Thus,

$$\begin{aligned}\mathcal{L}[f(t-a)u(t-a)] &= \int_{0^-}^{\infty} f(x)e^{-s(x+a)} dx \\ &= e^{-as} \int_{0^-}^{\infty} f(x)e^{-sx} dx = e^{-as} F(s)\end{aligned}$$

$$\mathcal{L}[f(t-a)u(t-a)] = e^{-as} F(s)$$

If a function is delayed in time by a , the result in the s -domain is found by multiplying the Laplace transform of the function (without the delay) by e^{-as} . This is called the **time-delay** or **time shift property** of the Laplace transform.

Laplace Transform

Frequency Shift: If $F(s)$ is the Laplace transform of $f(t)$, then

$$\begin{aligned}\mathcal{L}[e^{-at} f(t) u(t)] &= \int_0^{\infty} e^{-at} f(t) e^{-st} dt \\ &= \int_0^{\infty} f(t) e^{-(s+a)t} dt = F(s + a)\end{aligned}$$

$$\mathcal{L}[e^{-at} f(t) u(t)] = F(s + a)$$

The Laplace transform of $e^{-at} f(t)$ can be obtained from the Laplace transform of $f(t)$ by replacing every s with $s + a$. This is known as **frequency shift** or **frequency translation**.

Time Differentiation: Given that $F(s)$ is the Laplace transform of $f(t)$, the Laplace transform of its derivative is

$$\mathcal{L}\left[\frac{df}{dt} u(t)\right] = \int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt$$

Laplace Transform - Properties

To integrate this by parts, we let $u = e^{-st}$, $du = -se^{-st} dt$, and $dv = (df/dt) dt = df(t)$, $v = f(t)$. Then

$$\begin{aligned}\mathcal{L}\left[\frac{df}{dt}u(t)\right] &= f(t)e^{-st}\Big|_{0^-}^{\infty} - \int_{0^-}^{\infty} f(t)[-se^{-st}] dt \\ &= 0 - f(0^-) + s \int_{0^-}^{\infty} f(t)e^{-st} dt = sF(s) - f(0^-)\end{aligned}$$

$$\mathcal{L}[f'(t)] = sF(s) - f(0^-)$$

The Laplace transform of the second derivative of $f(t)$ is a repeated application

$$\begin{aligned}\mathcal{L}\left[\frac{d^2f}{dt^2}\right] &= s\mathcal{L}[f'(t)] - f'(0^-) = s[sF(s) - f(0^-)] - f'(0^-) \\ &= s^2F(s) - sf(0^-) - f'(0^-)\end{aligned}$$

$$\mathcal{L}[f''(t)] = s^2F(s) - sf(0^-) - f'(0^-)$$

$$\begin{aligned}\mathcal{L}\left[\frac{d^nf}{dt^n}\right] &= s^nF(s) - s^{n-1}f(0^-) \\ &\quad - s^{n-2}f'(0^-) - \cdots - s^0f^{(n-1)}(0^-)\end{aligned}$$

Laplace Transform - Properties

Time Integration: If $F(s)$ is the Laplace transform of $f(t)$, the Laplace transform of its integral is

$$\mathcal{L}\left[\int_0^t f(x) dx\right] = \int_{0^-}^{\infty} \left[\int_0^t f(x) dx\right] e^{-st} dt$$

To integrate this by parts, we let

$$u = \int_0^t f(x) dx, \quad du = f(t) dt$$

and

$$dv = e^{-st} dt, \quad v = -\frac{1}{s} e^{-st}$$

Then

$$\begin{aligned} \mathcal{L}\left[\int_0^t f(x) dx\right] &= \left[\int_0^t f(x) dx\right] \left(-\frac{1}{s} e^{-st}\right) \Big|_{0^-}^{\infty} \\ &\quad - \int_{0^-}^{\infty} \left(-\frac{1}{s}\right) e^{-st} f(t) dt \end{aligned}$$

$$\mathcal{L}\left[\int_0^t f(x) dx\right] = \frac{1}{s} F(s)$$

Laplace Transform - Properties

Frequency Differentiation: If $F(s)$ is the Laplace transform of $f(t)$, then

If $F(s)$ is the Laplace transform of $f(t)$, then

$$F(s) = \int_{0^-}^{\infty} f(t)e^{-st} dt$$

Taking the derivative with respect to s ,

$$\frac{dF(s)}{ds} = \int_{0^-}^{\infty} f(t)(-te^{-st}) dt = \int_{0^-}^{\infty} (-tf(t))e^{-st} dt = \mathcal{L}[-tf(t)]$$

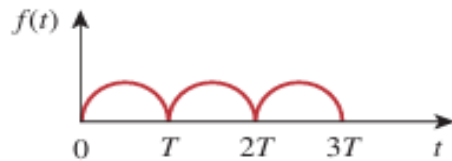
and the frequency differentiation property becomes

$$\mathcal{L}[tf(t)] = -\frac{dF(s)}{ds}$$

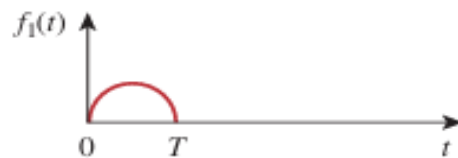
$$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n F(s)}{ds^n}$$

Laplace Transform - Properties

Time Periodicity: If function $f(t)$ is a periodic function such as shown, it can be represented as the sum of time-shifted functions

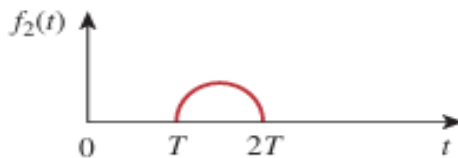


$$\begin{aligned} f(t) &= f_1(t) + f_2(t) + f_3(t) + \dots \\ &= f_1(t) + f_1(t - T)u(t - T) \\ &\quad + f_1(t - 2T)u(t - 2T) + \dots \end{aligned}$$

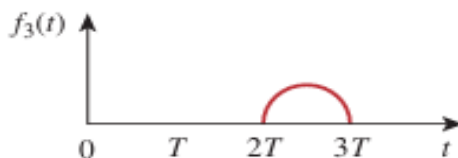


$$f_1(t) = \begin{cases} f(t), & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$$

We now transform each term in the equation and apply the time-shift property.



$$\begin{aligned} F(s) &= F_1(s) + F_1(s)e^{-Ts} + F_1(s)e^{-2Ts} + F_1(s)e^{-3Ts} + \dots \\ &= F_1(s)[1 + e^{-Ts} + e^{-2Ts} + e^{-3Ts} + \dots] \end{aligned}$$



$$1 + x + x^2 + x^3 + \dots = \frac{1}{1 - x}$$

Laplace transform of a periodic function is the transform of the first period of the function divided by $1 - e^{-Ts}$.

$$F(s) = \frac{F_1(s)}{1 - e^{-Ts}}$$

Laplace Transform-Properties

Property	$f(t)$	$F(s)$
Linearity	$a_1 f_1(t) + a_2 f_2(t)$	$a_1 F_1(s) + a_2 F_2(s)$
Scaling	$f(at)$	$\frac{1}{a} F\left(\frac{s}{a}\right)$
Time shift	$f(t - a)u(t - a)$	$e^{-as} F(s)$
Frequency shift	$e^{-at} f(t)$	$F(s + a)$
Time differentiation	$\frac{df}{dt}$	$sF(s) - f(0^-)$
	$\frac{d^2 f}{dt^2}$	$s^2 F(s) - sf(0^-) - f'(0^-)$
	$\frac{d^3 f}{dt^3}$	$s^3 F(s) - s^2 f(0^-) - sf'(0^-) - f''(0^-)$
	$\frac{d^n f}{dt^n}$	$s^n F(s) - s^{n-1} f(0^-) - s^{n-2} f'(0^-) - \dots - f^{(n-1)}(0^-)$
Time integration	$\int_0^t f(x) dx$	$\frac{1}{s} F(s)$
Frequency differentiation	$tf(t)$	$-\frac{d}{ds} F(s)$

Laplace Transform - Properties

Property	$f(t)$	$F(s)$
Frequency differentiation	$tf(t)$	$-\frac{d}{ds}F(s)$
Frequency integration	$\frac{f(t)}{t}$	$\int_s^\infty F(s)ds$
Time periodicity	$f(t) = f(t + nT)$	$\frac{F_1(s)}{1 - e^{-sT}}$
Initial value	$f(0)$	$\lim_{s \rightarrow \infty} sF(s)$
Final value	$f(\infty)$	$\lim_{s \rightarrow 0} sF(s)$
Convolution	$f_1(t) * f_2(t)$	$F_1(s)F_2(s)$

Laplace Transform - initial value and final value

The initial-value and final-value properties allow us to find the initial value $f(0)$ and the final value $f(\infty)$ of $f(t)$ directly from its Laplace transform $F(s)$.

To obtain these properties, we begin with the differentiation property,

$$sF(s) - f(0) = \mathcal{L}\left[\frac{df}{dt}\right] = \int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt \quad \text{Eqn.1}$$

If we let s tending to ∞ , the integrand vanishes.

$$\lim_{s \rightarrow \infty} [sF(s) - f(0)] = 0$$

$f(0)$ is independent of s

$$f(0) = \lim_{s \rightarrow \infty} sF(s)$$

This is the **initial value theorem**.

In Eqn.1, if we let s tending to 0, then

$$\lim_{s \rightarrow 0} [sF(s) - f(0^-)] = \int_{0^-}^{\infty} \frac{df}{dt} e^{0t} dt = \int_{0^-}^{\infty} df = f(\infty) - f(0^-)$$

$$f(\infty) = \lim_{s \rightarrow 0} sF(s)$$

This is the **final value theorem**.

Laplace Transform

1. Obtain the Laplace transform of $f(t) = \delta(t) + 2u(t) - 3e^{-2t}u(t)$

$$\begin{aligned} F(s) &= \mathcal{L}[\delta(t)] + 2\mathcal{L}[u(t)] - 3\mathcal{L}[e^{-2t}u(t)] \\ &= 1 + 2\frac{1}{s} - 3\frac{1}{s+2} = \frac{s^2 + s + 4}{s(s+2)} \end{aligned}$$

2. Obtain the Laplace transform of $f(t) = (\cos(2t) + e^{-4t})u(t)$.

Answer: $\frac{2s^2 + 4s + 4}{(s+4)(s^2+4)}$.

3. Obtain the Laplace transform of $f(t) = t^2 \sin 2t u(t)$.

$$\mathcal{L}[\sin 2t] = \frac{2}{s^2 + 2^2}$$

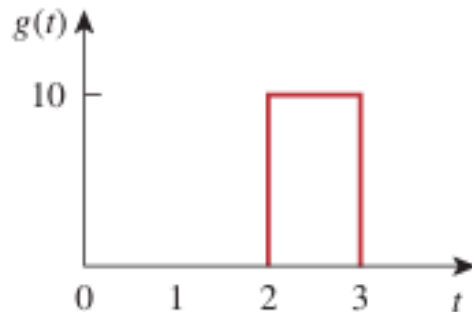
Using frequency differentiation property,
$$\begin{aligned} F(s) = \mathcal{L}[t^2 \sin 2t] &= (-1)^2 \frac{d^2}{ds^2} \left(\frac{2}{s^2 + 4} \right) \\ &= \frac{d}{ds} \left(\frac{-4s}{(s^2 + 4)^2} \right) = \frac{12s^2 - 16}{(s^2 + 4)^3} \end{aligned}$$

Laplace Transform

4. Find the Laplace transform of $f(t) = t^2 \cos 3t u(t)$

Answer: $\frac{2s(s^2 - 27)}{(s^2 + 9)^3}$.

5. Find the Laplace transform of the gate function shown in the figure below.



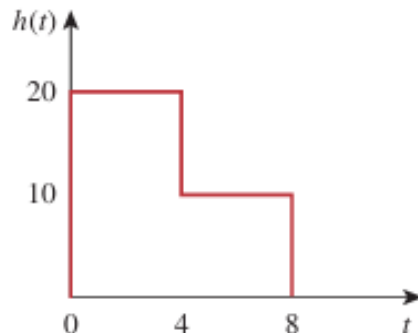
We can express this gate function as,

$$g(t) = 10[u(t - 2) - u(t - 3)]$$

apply the time-shift property to get its Laplace transform.

$$G(s) = 10\left(\frac{e^{-2s}}{s} - \frac{e^{-3s}}{s}\right) = \frac{10}{s}(e^{-2s} - e^{-3s})$$

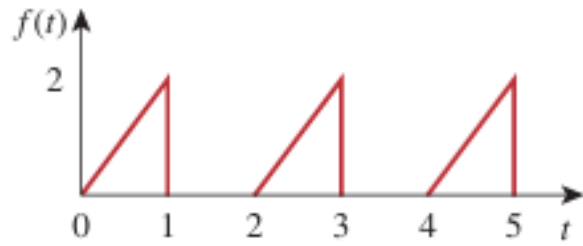
6. Find the Laplace transform of the function $h(t)$ given in the figure.



Answer: $\frac{5}{s}(2 - e^{-4s} - e^{-8s})$.

Laplace Transform

7. Calculate the Laplace transform of the periodic function shown in the figure below.



The period of the function is $T = 2$. Use the property of Time periodicity.

$$\begin{aligned} f_1(t) &= 2t[u(t) - u(t - 1)] = 2tu(t) - 2tu(t - 1) \\ &= 2tu(t) - 2(t - 1 + 1)u(t - 1) \\ &= 2tu(t) - 2(t - 1)u(t - 1) - 2u(t - 1) \end{aligned}$$

Using the time shift property, we get,

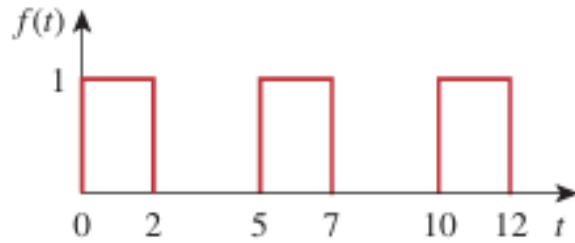
$$F_1(s) = \frac{2}{s^2} - 2\frac{e^{-s}}{s^2} - \frac{2}{s}e^{-s} = \frac{2}{s^2}(1 - e^{-s} - se^{-s})$$

Therefore the transform of the periodic function is,

$$F(s) = \frac{F_1(s)}{1 - e^{-Ts}} = \frac{2}{s^2(1 - e^{-2s})}(1 - e^{-s} - se^{-s})$$

Laplace Transform

8. Determine the Laplace transform of the periodic function given below.



Answer: $\frac{1 - e^{-2s}}{s(1 - e^{-5s})}$

Next class: Laplace Transform